

DSA



FULL COURSE

Topics



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DSA Roadmap

START

Foundations

- Programming Basics (Python, C++)
- Basic Mathematical Concepts

Data Structure

- Array & Lists
- Linked Lists
- Stacks & Queues

Searching & Sorting

- Linear & Binary search
- Graph representation

Advanced Data Structures

- Hashing
- Heaps
- Hash maps & Sets

Trees & Graph

- Binary Trees
- Graph Representation

Recursion

- Understanding Recursion
- Recursive Algorithms

Algorithms

- Greedy Algorithms
- Divide and Conquer
- Dynamic Programming

Advanced Topics

- Backtracking
- String Algorithms
- Bit Manipulation

Problem Solving

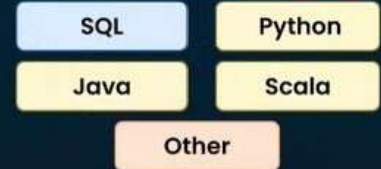
- Practice on Platforms (LeetCode)
- Solve Algorithmic Challenges

Projects & Portfolio

- Build Real-World Projects
- Showcase Skills on GitHub

Data Engineering Burger

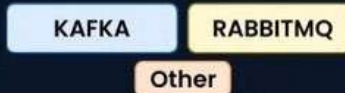
01. Learn Programming



02. Processing



04. Message Queue



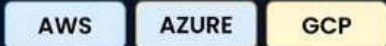
03. Databases



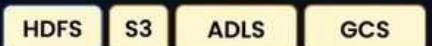
06. Warehouse



05. Cloud Computing



07. Storage



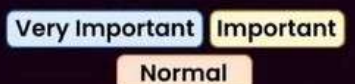
08. Data Lake



10. Resource Manager



09. Orchestration



Tips for Learning DSA

1. Practice Regularly

- Solve problems on competitive programming sites

2. Understand Concepts Thoroughly

- Focus on understanding rather than memorizing

3. Build Projects

- Apply concepts in real-world projects

4. Collaborate and Discuss

- Join study groups or online forums



Module 1: Introduction to Data Structures and Algorithms

1. Introduction to DSA

- Definition and importance
- Real-world applications

2. Algorithm Analysis

- Time complexity
- Space complexity
- Big O notation

Module 2: Basic Data Structures

1. Arrays

- Definition and operations
- Applications and examples

2. Linked Lists

- Singly linked list
- Doubly linked list
- Circular linked list

3. Stacks

- Definition and operations
- Applications

4. Queues

- Definition and operations
- Circular queue
- Priority queue



Module 3: Advanced Data Structures

1. Trees

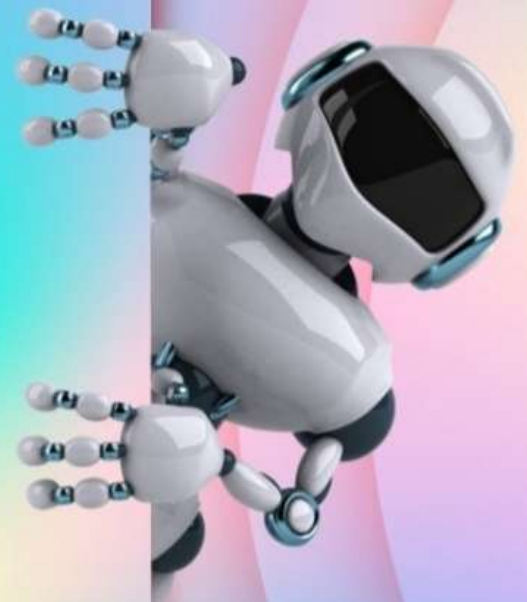
- Binary trees
- Binary search trees
- AVL trees
- Red-Black trees

2. Heaps

- Min-Heap
- Max-Heap
- Heap operations

3. Graphs

- Graph representation (adjacency matrix and list)
- Graph traversal (BFS, DFS)
- Shortest path algorithms (Dijkstra, Bellman-Ford)



Module 4: Sorting and Searching Algorithms

1. Sorting Algorithms

- Bubble sort
- Selection sort
- Insertion sort
- Merge sort
- Quick sort
- Heap sort

2. Searching Algorithms

- Linear search
- Binary search
- Interpolation search

Module 5: Advanced Algorithms

1. Dynamic Programming

- Introduction and concepts
- Fibonacci sequence
- Knapsack problem
- Longest common subsequence

2. Greedy Algorithms

- Introduction and concepts
- Huffman coding
- Kruskal's algorithm
- Prim's algorithm

3. Backtracking

- Introduction and concepts
- N-Queens problem
- Rat in a maze



Module 6: String Algorithms

1. Pattern Matching

- Naive algorithm
- KMP algorithm
- Rabin-Karp algorithm

2. Tries

- Introduction and operations
- Applications

Module 7: Miscellaneous Topics

1. Bit Manipulation

- Basics and operations
- Applications

2. Divide and Conquer

- Concept and applications

3. Complex Data Structures

- Segment trees
- Fenwick trees

4. Graph Algorithms

- Topological sorting
- Floyd-Warshall algorithm
- Bellman-Ford algorithm



Module 8: Practical Applications and Problem Solving

1. Case Studies

- Real-world problems and their solutions

2. Competitive Programming

- Techniques and tips
- Practice problems

Module 9: Project

1. Course Project

- Implement a complex real-world problem using the learned data structures and algorithms.

Resources and References

1. Books

- "Introduction to Algorithms" by Cormen, Leiserson, Rivest, and Stein
- "Algorithms" by Robert Sedgewick and Kevin Wayne

2. Online Resources

- GeeksforGeeks
- LeetCode
- HackerRank
- Coursera, Udemy, and other MOOC platforms

